# Huy Huong

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## Education

**University of Maryland, College Park, MD** — **Expected May 2027**
M.S. Aerospace Engineering — GPA: 3.740

**University of Bridgeport, Bridgeport, CT** — **May 2025**
B.S. Computer Science & Mathematics, Minor in Physics, Honors Program — GPA: 3.679

## Technical Skills

**Programming:** Python, C++, C, Java, ROS2, MATLAB, Git, Linux

**Engineering:** Underwater SLAM, High Altitude Ballooning, Robotics, Live Telemetry

**CAD/Design:** SolidWorks, AutoCAD, KiCad, Eagle, OnShape

**Manufacturing:** CNC Machining, 3D Printing, Laser Cutting

## Research Experience

**Naval Research Laboratory (NRL)** — Washington D.C.
*NREIP Intern, Laboratory for Autonomous Systems Research (5517)* — Jun 2026 to Aug 2026

- Analyzed ERA5 and radiosonde atmospheric data to support research on optimal station-keeping locations for solar high-altitude balloons (sHABs)
- Redesigned the forecast processing pipeline by unifying ERA5 and GFS workflows, adding multi-year data support, and improving pipeline reliability through resumable multiprocessing in Python
- Increased the accuracy of wind analysis by optimizing interpolation methods and updating HovMoller wind-direction processing, while developing tools to validate and migrate historical datasets
- Programmed and maintained solar high-altitude balloon systems for upcoming research flights and atmospheric data collection

**Nationwide Eclipse Ballooning Project (NEBP)**

*Mentor* — Jun 2026 to Present

- Mentored 7 undergraduate students through payload design, integration, and flight readiness for high-altitude balloon launches during the August 12, 2026 total solar eclipse in Spain
- Expanded the CHASE tracking platform with Spanish airspace overlays (control zones, terminal areas, restricted and military zones) to support launch site selection and landing prediction for international flights
- Advised teams on Spanish airspace regulations and ENAIRE coordination requirements for unmanned free balloon launches

*Student Pod Lead* — Sep 2022 to Nov 2024

- Successfully collected solar eclipse data under extreme conditions by leading 10 engineers in designing resilient payload systems that withstood 100+ MPH winds and altitudes exceeding 100,000 feet during Annular and Total eclipse missions
- Achieved 100% milestone completion for multi-regional eclipse project by supervising cross-functional student and faculty teams across the East Central Region, resolving critical hardware and software integration issues
- Coordinated and analyzed eclipse data across 53 teams during 2023-2024 campaigns, implementing quality control processes that enabled successful publication on Dryad scientific repository (Des Jardins, et al. 2025)
- National Aeronautics and Space Administration, Subaward Title: Montana Space Grant Consortium Opportunities in STEM 2020-2024, Subaward ID: G355-22-W9649

**University of Maryland Motion and Teaming Laboratory** College Park, MD
*Graduate Research Assistant* Aug 2025 to Present

- Analyzing Underwater SLAM applications for multi-vehicle navigation systems and current methodologies and identifying research gaps for autonomous underwater vehicle coordination
- Designed a stigmergy-based data sharing protocol using k-means clustered markers, allowing sequential AUVs to retrieve, merge, and update prior map data without direct communication or centralized storage

**Exploring Naval Underwater STEM (Exploring NUSTEM)** Bridgeport, CT
*Undergraduate Researcher* Dec 2023 to Aug 2025

- Resolved critical corrosion issues by conducting oscilloscope diagnostics, systematically treating corroded submarine components, and applying protective rust-resistant coatings for "The Explorer" ROV conversion
- Enhanced deep-sea exploration by constructing and modifying three BlueROV 2 kits with high-precision sonar, achieving successful data collection at 300-meter depths in Long Island Sound
- Led design reviews with industry professionals during Explorer submarine restoration, ensuring compliance with operational standards and safety requirements for deployment readiness
- Office of Naval Research (ONR) FOA N00014-21-S-F004 Science, Technology, Engineering & Mathematics (STEM), Education and Workforce Program, "Exploring Naval Underwater STEM (Exploring NUSTEM)"

**Great Lunar Expedition for Everyone (GLEE)** Bridgeport, CT
*Team 2149 Lead* Apr 2022 to Sep 2023

- Led an interdisciplinary team of 10 to design, test, and prepare for the launch of "LunaSat" payloads, which were engineered to conduct localized scientific missions on the lunar surface
- Facilitated team meetings and cross-functional collaboration, improving communication and ensuring efficient problem-solving throughout the project

## Projects

**CHASE (HAB Tracking & Recovery Map)**

- Extended Project Horus' open-source chasemapper into a Python and JavaScript web mapping platform used by a 100+ member organization to coordinate launch-day tracking and recovery, unifying APRS-IS and SPOT Trace telemetry with live landing predictions
- Developed HTML/JavaScript overlays for FAA airspace, temporary flight restrictions, and Maryland parcel data, enabling teams to check restrictions and contact landowners at predicted landing sites
- Containerized the Flask backend with Docker, continuous integration, and automated tests for headless Raspberry Pi deployment in the field

**High Altitude Robotic Monkey (HAM)**

- Overhauled robotic monkey design for high-altitude balloon flights, implementing live streaming and ground station interaction capabilities to enable real-time K-12 STEM education experiences with space-based robotics
- Designed and manufactured 5 custom PCBs for payload system integration while redesigning serial communication protocols, reducing dropped packets by 30% and decreasing overall payload weight by 35%

**Parrot Oriented Logical Interface (POLI)**

- Developed English & Spanish speech recognition engine in C++ using Nano BLE Sense microphone, achieving real-time transcription capabilities and enhancing interactive response time by 20%
- Designed detailed 3D parrot model in SolidWorks with precise geometrical constraints, enabling accurate prototype visualization that improved stakeholder approval and project funding

## Employment

**Aerospace Engineering Department, University of Maryland** College Park, MD
*Graduate Teaching Assistant* Jan 2026 to Present

- **Courses:** ENAE202 Computing Fundamentals for Engineers (Spring 2026), ENAE380 Flight System Softwares (Fall 2026)

- Lead weekly office hours for 60+ students to review course material and homework
- Grade assignments and provide thorough, timely feedback for 190 students across 6 sections

**University of Bridgeport STEM on Wheels** Bridgeport, CT
*STEM Instructor* May 2023 to Jul 2025

- Reached over 8,000 students through mobile STEM outreach, delivering hands-on robotics, coding, and engineering lessons to 100+ students per visit
- Enhanced student comprehension of robotics, coding, and engineering through immersive mobile laboratory experiences, sparking sustained interest in technical fields
- Maximized learning outcomes by adapting STEM curriculum across K-12 grade levels, enabling successful participation in advanced experiments

**University of Bridgeport Extreme Environments Laboratory** Bridgeport, CT
*Undergraduate Researcher* Sep 2022 to Aug 2025

- Expanded STEM access to underserved communities through NSF, U.S. Naval Department of Education, and NASA-funded outreach programs, delivering hands-on materials and activities
- Validated material and system performance for aerospace and deep-sea applications through comprehensive testing under extreme environmental conditions

## Published Conference Proceedings

[4] **H. T. Huong** and M. W. Otte. "Utility-Driven D*Lite for Underwater Multi-Agent Exploration through Stigmergy". *Workshop on Perception and Navigation for Marine Robotics at IROS 2026*. Pittsburgh, PA, Sept. 2026. URL: https://openreview.net/forum?id=vVl5b6KkjZ.

[3] H. Gu, T. Ibrahim, **H. T. Huong**, S. Gudi, R. Mitra, X. Zhang, N. Zoghb, J. Pallis, and C. Qiu. "Explore Atmospheric Transformation of Marine Bioaerosol via a High-Altitude Ballooning Platform". In: *Academic High Altitude Conference* 2026.1 (May 2026). DOI: 10.31274/ahac.25535. URL: https://doi.org/10.31274/ahac.25535.

[2] **H. T. Huong**, S. Gudi, V. Sripada, J. D. Urrea Vargas, M. O. AL-Torzi, K. Bird, and J. M. Pallis. "High Altitude Robotic Monkey". *2025 ASEE Annual Conference & Exposition*. Montreal, Quebec, Canada: ASEE Conferences, June 2025. DOI: 10.18260/1-2--56690. URL: https://peer.asee.org/56690.

[1] V. Sripada, M. O. AL-Torzi, **H. T. Huong**, J. D. Urrea Vargas, S. Gudi, A. L. Iorio, J. M. Pallis, and J. McNitt-Gray. "Modification of a 1-Person Submarine for Remote Control Operation". *2025 ASEE Annual Conference & Exposition*. Montreal, Quebec, Canada: ASEE Conferences, June 2025. DOI: 10.18260/1-2--56974. URL: https://peer.asee.org/56974.

## Refereed Posters

[6] **H. Huong**, S. Gudi, R. Mitra, J. Urrea Vargas, M. Al-Torzi, D. Howard, U. Sfregoli, S. Goodman, K. Bird, A. Edward, N. Belfiore, N. Zoghb, and J. Pallis. "Nationwide Ballooning Project - Solar Max Radiation Project". *2026 ASEE Northeast Section Conference*. Manchester, NH, Mar. 2026.

[5] V. Sripada, J. Urrea Vargas, P. Suarez, M. O. Al-Torzi, **H. Huong**, J. Pallis, and N. Zoghb. "Transforming a One-Person, One-Atmosphere Submarine into an Autonomous Underwater Vehicle". *2026 ASEE Northeast Section Conference*. Manchester, NH, Mar. 2026.

[4] **H. Huong**, M. O. Al-Torzi, J. Urrea Vargas, R. Mitra, N. Zoghb, and J. Pallis. "University of Bridgeport High-Altitude Ballooning Program - Past and Future". *2025 Scientific Ballooning Technologies Workshop*. Minneapolis, MN, May 2025.

[3] **H. Huong**, J. Medina, A. Armatis, C. Lara, and S. Patel. "LawnBot: Affordable Autonomous Lawnmowers". *2025 ASEE Northeast Section Conference*. Bridgeport, CT, Mar. 2025.

[2] **H. Huong**, J. Urrea Vargas, V. Sripada, D. Howard, J. Pallis, D. Mestre, L. Reed, G. Moyher, and S. Zhang. "High Altitude Monkey 2.0". *Between Eclipse Research Conference (BERC 2024)*. Virtual, Jan. 2024.

[1] **H. Huong**, J. Urrea Vargas, V. Sripada, A. Saoudi, S. Gudi, D. Howard, R. Beadle, J. Pallis, D. Mestre, L. Reed, G. Moyher, and S. Zhang. "High Altitude Ballooning During the 2023 Annular Solar Eclipse". *Between Eclipse Research Conference (BERC 2024)*. Virtual, Jan. 2024.

## Leadership & Service

**University of Maryland Balloon Payload Program** Aug 2025 to Present
*VIP-208E/308E/408E Co-Instructor - Fall 2025*

- Supported five undergraduate engineering teams in the design, fabrication, and qualification of high-altitude scientific payloads, guiding requirements development, subsystem integration, and design reviews.
- Provided instruction on data acquisition workflows, sensor calibration, flight testing, and post-flight analysis to prepare students for advanced research and independent project work.

*Freshman Workshop Mentor (FLOATER)*

- Guided a team of first year students through the end-to-end development of the FLOATER high-altitude payload, including subsystem definition, requirements flowdown, and design reviews.
- Instructed students in core engineering workflows & surface-mount soldering, PCB layout (KiCad), parametric CAD modeling (Onshape/Fusion), and iterative prototyping with rapid design & test cycles.

**American Society for Engineering Education (ASEE)** Jul 2025 to Present
*Aerospace Division Committee - Student Member-At-Large*

- Serving on ASEE Aerospace Division Committee to review and evaluate aerospace engineering research abstracts and papers, providing technical feedback for annual convention

**University of Bridgeport Aerospace Club** Apr 2022 to May 2025
*President*

- Led aerospace club activities including technical workshops and industry guest lectures, promoting hands-on engineering education and collaboration across multiple disciplines
- Organized community stargazing events for students, faculty, and local residents, providing hands-on astronomy education and constellation observation experiences

**Student Government Association**, University of Bridgeport, Bridgeport, CT Sep 2022 to May 2025
*Member*

- Worked with clubs and communities across campus to initiate programming for the entire student body.

## Honors & Awards

- Academic Achievement Award for BS. Computer Science Recipient, May 2025
- UB RISE Conference Undergraduate Poster Category - Third Place Recipient: “LawnBot: Affordable Autonomous Lawnmower”, Apr 2025
- University of Bridgeport Outstanding Involvement by a Junior Award Recipient, Apr 2024
- Upsilon Pi Epsilon International Honors Society: Delta Chapter Member, Spring 2022 - Spring 2025
- Connecticut Space Grant Consortium Undergraduate Scholarship Recipient, Fall 2022
- University of Bridgeport Honors Program, Fall 2021 - Spring 2025
- University of Bridgeport College of Engineering Dean’s List, Fall 2021 - Spring 2025